Bagged Random Subset Methods for Forecast Combination under Common Loading

Theory, Optimal Subset Size, Dominance Regions, and Empirical Applications

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Abstract

We study Random Subset Methods (RSM) for forecast combination in the common-loading heteroskedastic forecast-error model. The paper makes four contributions. First, we derive an exact representation of the infinite-bagged population risk $Q_k^{\text{bag},\infty}$: it equals the oracle optimum plus a weighted squared distance between the average RSM weight and the oracle weight. Under small precision heterogeneity this distance admits a closed-form small-heterogeneity approximation $C_\delta(1/k - 1/m)^2$, which under a standard finite-sample inflation approximation yields a cube-root optimal subset size $k^* \propto T^{1/3}$—a lower scaling exponent than the square-root rule of the average-subset estimator. Under general heterogeneity the excess risk follows a power law $R_k \approx C z_k^\alpha$ with $\alpha \in [1, 2]$, giving $k^* \propto T^{1/(\alpha+1)}$. Second, we derive exact and approximate dominance conditions under which RSM strictly beats the equal-weighted mean and full estimated optimal weights. Third, Monte Carlo experiments in both the exogenous (known Σ_e) and endogenous (estimated Σ_e) settings confirm these predictions. The endogenous k^* is substantially higher than the exogenous prediction, consistent with the extended estimation-noise interpretation. Fourth, applying RSM to FRED-MD GDP forecasts yields statistically significant gains over the equal-weighted mean (RMSFE reduction of 10–30%), while the Survey of Professional Forecasters—a near-homogeneous panel with negligible precision heterogeneity ($C_\delta \approx 0$)—confirms that RSM offers little advantage over the simple mean when forecasters are nearly identical in quality.

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1 Scope and Main Message

The central question is: in a large panel of m individual forecasters, how many should a practitioner randomly subsample, and how should subsample forecasts be combined, to minimise out-of-sample loss?

Two distinct RSM objects have been studied in the literature. The *average-subset estimator* picks a random size-k subset, fits constrained OLS within it, and reports that single forecast. The *bagged estimator* averages G such draws. Since $w \mapsto w'\Sigma_e w$ is convex, Jensen's inequality gives

$$Q_k^{\text{bag},\infty} \;\leq\; Q_k^{\text{sub}}, \tag{1}$$

so bagging always weakly reduces population risk. Yet prior work has largely analysed Q_k^{sub} and assumed a functional form $Q_k^{\text{bag},\infty} - Q^{\text{opt}} \approx C/k^2$. We replace this assumption with a derivation.

Main message. Under small precision heterogeneity, the endpoint-corrected approximation

$$Q_k^{\text{bag},\infty} - Q^{\text{opt}} \;\approx\; C_\delta\left(\frac{1}{k} - \frac{1}{m}\right)^2, \qquad C_\delta = \frac{BV_\delta}{m}\left(\frac{m}{m-1}\right)^2, \quad B = \frac{1}{\bar{\tau}}, \tag{2}$$

follows from the bagged weight expansion, respects the exact boundary condition $Q_m^{\text{bag},\infty} = Q^{\text{opt}}$, and implies a cube-root optimal subset size $k^* \propto T^{1/3}$. Under general heterogeneity, a power-law slope $\hat{\alpha} \in [1,2]$ governs the decay of excess risk and determines the T-scaling exponent $1/(\alpha+1)$.

2 Forecast-Combination Setup

Assumption 1 (Forecast model)**.** *There are m individual forecasts $f_t = (f_{1t}, \ldots, f_{mt})'$ of a scalar target $y_t = \mu_t + \varepsilon_t$. Forecast errors relative to the predictable component are $u_t = f_t - \mu_t \iota_m$. Assume*

$$E(\varepsilon_t) = 0, \quad \operatorname{Var}(\varepsilon_t) = s > 0, \quad E(\varepsilon_t u_{it}) = 0, \quad E(u_t) = 0, \quad \operatorname{Var}(u_t) = \Sigma_e \succ 0.$$

For any weight vector w with $w'\iota_m = 1$,

$$w'f_t - y_t = w'u_t - \varepsilon_t, \qquad E\big[(w'f_t - y_t)^2\big] = s + w'\Sigma_e w. \tag{3}$$

The common innovation variance s is shared by all combinations, so population differences are governed by

$$Q(w) = w'\Sigma_e w. \tag{4}$$

The total population loss is $L(w) = s + Q(w)$.

3 Benchmarks

3.1 Equal-Weighted Mean

The equal-weighted (EW) combination uses

$$w^{\text{ave}} = \frac{1}{m}\iota_m, \qquad Q^{\text{ave}} = \frac{1}{m^2}\iota_m'\Sigma_e\iota_m. \tag{5}$$

Since no weights are estimated, there is no finite-sample inflation and the loss is simply $L^{\text{ave}} = s + Q^{\text{ave}}$.

3.2 Full Optimal Weights

The infeasible minimum-variance combination solves $\min_w w'\Sigma_e w$ subject to $w'\iota_m = 1$. Its solution and population risk are

$$w^{\text{opt}} = \frac{\Sigma_e^{-1}\iota_m}{\iota_m'\Sigma_e^{-1}\iota_m}, \qquad Q^{\text{opt}} = \left(\iota_m'\Sigma_e^{-1}\iota_m\right)^{-1}. \tag{6}$$

When Σ_e is estimated from T observations, the degrees-of-freedom approximation gives the feasible optimal loss

$$L^{\text{opt}} \approx (s + Q^{\text{opt}})\frac{T-1}{T-m}, \qquad T > m. \tag{7}$$

This loss diverges as $m \to T$, reflecting the curse of dimensionality when the number of parameters approaches the sample size.

3.3 EW and Full Optimal as Limiting Cases of RSM

Under the common-loading model (Section 6), the RSM estimator unifies both benchmarks:

- *Homogeneous case* ($\tau_i = \bar{\tau}$ for all i): weights within any subset are equal, so $\bar{v}_{k,i} = 1/m$ for all k, and $Q_k^{\text{bag},\infty} = Q^{\text{opt}} = Q^{\text{ave}}$. RSM and EW coincide in population.
- *Full pool* ($k = m$): the only subset is $\{1, \dots, m\}$, so $w_S = w^{\text{opt}}$ and $Q_m^{\text{bag},\infty} = Q^{\text{opt}}$. RSM recovers the full optimal at $k = m$.

For $1 \le k < m$ and $V_\delta > 0$, RSM interpolates between these limits with strictly positive excess risk $R_k^{\text{bag},\infty} > 0$.

4 RSM for Forecast Combination

Definition 1 (RSM)**.** Let $S \subset \{1, \dots, m\}$ be a random subset of size k, drawn uniformly without replacement. Let $\Sigma_{e,S}$ be the $k \times k$ principal submatrix of Σ_e indexed by S. The *optimal subset weight vector* is

$$w_S = \frac{\Sigma_{e,S}^{-1} \iota_S}{\iota_S' \Sigma_{e,S}^{-1} \iota_S}, \tag{8}$$

embedded into $\mathbb{R}^m$ as $v_S = P_S w_S$ where P_S places the k selected weights in their original coordinates and sets remaining coordinates to zero.

4.1 Average-Subset RSM

The average-subset estimator draws a single subset S and reports the corresponding forecast. Its population risk is

$$Q_k^{\text{sub}} = E_S[v_S' \Sigma_e v_S] = E_S[Q(S)], \qquad Q(S) = \left(\iota_S' \Sigma_{e,S}^{-1} \iota_S\right)^{-1}. \tag{9}$$

4.2 Infinite-Bagged RSM

Let G independent subsets $S_1, \dots, S_G$ be drawn and define $\hat{v}_{k,G} = G^{-1} \sum_{g=1}^{G} v_{S_g}$. The population risk of this bagged forecast is $Q_k^{\text{bag},G} = E[\hat{v}_{k,G}' \Sigma_e \hat{v}_{k,G}]$.

The *infinite-bagged weight vector* is

$$\bar{v}_k = E_S[v_S], \tag{10}$$

and the *infinite-bagged population risk* is

$$Q_k^{\text{bag},\infty} = \bar{v}_k' \Sigma_e \bar{v}_k. \tag{11}$$

4.3 Finite-Bagging Bridge

Proposition 1 (Finite-bagging bridge)**.** *For G independent subset draws,*

$$Q_k^{\text{bag},G} = Q_k^{\text{bag},\infty} + \frac{1}{G}\left(Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}\right). \tag{12}$$

Consequently, $Q_k^{\text{bag},\infty} \le Q_k^{\text{bag},G} \le Q_k^{\text{sub}}$ for all $G \ge 1$.

Proof. Let $V = v_S$ and $\bar{v} = E[V]$. For G independent draws, $\hat{v}_{k,G} = G^{-1} \sum_g V_g$. Then

$$\begin{aligned} E[\hat{v}_{k,G}' \Sigma_e \hat{v}_{k,G}] &= \bar{v}' \Sigma_e \bar{v} + E[(\hat{v}_{k,G} - \bar{v})' \Sigma_e (\hat{v}_{k,G} - \bar{v})] \\ &= Q_k^{\text{bag},\infty} + \frac{1}{G} E[(V - \bar{v})' \Sigma_e (V - \bar{v})] = Q_k^{\text{bag},\infty} + \frac{1}{G}(Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}). \end{aligned}$$

The inequalities follow from $\Sigma_e \succ 0$. $\square$

This identity is exact, with no approximation. It shows that infinite bagging removes the within-subset variance; finite bagging reintroduces a fraction $1/G$ of it.

5 Finite-Sample Inflation

For a single subset of size k, sum-to-one constrained OLS estimates $k-1$ free parameters from T observations. Under the Gaussian random-design approximation, the out-of-sample inflation factor is

$$I(T,k) = \frac{T-1}{T-k}. \tag{13}$$

The average-subset and bagged losses are therefore approximated by

$$L_k^{\mathrm{sub}} \approx (s + Q_k^{\mathrm{sub}})\frac{T-1}{T-k}, \qquad L_k^{\mathrm{bag}} \approx (s + Q_k^{\mathrm{bag}})\frac{T-1}{T-k}. \tag{14}$$

For bagged RSM the true effective degrees of freedom are smaller than k because averaging overlapping estimators reduces variance; the formula $I(T,k)$ is therefore a conservative upper bound for the bagged inflation. We use this conservative convention throughout. All results on optimal subset-size scaling (Section 9) are derived under this *maintained inflation approximation*; the cube-root rule should be read as a theoretical guide, not a precise empirical formula.

Remark 1 (Inflation and heterogeneity)**.** *The inflation factor $I(T,k) = (T-1)/(T-k)$ is increasing in k. It penalises large subsets, so the optimal k balances the approximation error (decreasing in k, since larger subsets approach the full optimum) against the estimation noise (increasing in k). This trade-off is the source of the interior optimum.*

6 Common-Loading Model: Exact Derivations

6.1 Model and Notation

Assumption 2 (Common-loading structure)**.** *The forecast-error covariance matrix takes the form*

$$\Sigma_e = q\iota_m\iota_m' + D, \qquad D = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_m^2), \tag{15}$$

where $q \geq 0$ is the common-loading variance and $\sigma_i^2 > 0$ are idiosyncratic variances. Define precision weights

$$\tau_i = \sigma_i^{-2}, \quad D_{ii} = \sigma_i^2 = \tau_i^{-1}, \quad A_S = \sum_{i\in S}\tau_i, \quad A_m = \sum_{i=1}^{m}\tau_i. \tag{16}$$

Under this model, for any w with $w'\iota_m = 1$,

$$Q(w) = w'\Sigma_e w = q + \sum_{i=1}^{m} \frac{w_i^2}{\tau_i}, \tag{17}$$

since $w'\iota_m = 1$ implies $q(w'\iota_m)^2 = q$. The common component q cannot be diversified away.

6.2 Exact Subset Risk

For subset S, $\Sigma_{e,S} = q\iota_S\iota_S' + D_S$. By the Sherman–Morrison formula,

$$\iota_S'\Sigma_{e,S}^{-1}\iota_S = A_S - \frac{qA_S^2}{1 + qA_S} = \frac{A_S}{1 + qA_S}. \tag{18}$$

Therefore

$$Q(S) = \left(\iota_S'\Sigma_{e,S}^{-1}\iota_S\right)^{-1} = q + \frac{1}{A_S}, \tag{19}$$

and the optimal subset weights are $w_{i,S} = \tau_i/A_S$ for $i \in S$. Hence

$$Q_k^{\text{sub}} = q + E_S\left[\frac{1}{A_S}\right]. \tag{20}$$

6.3 Full Optimal Risk and Weights

For the full pool, $A_m = \sum_{i=1}^m \tau_i$. From (19) with $S = \{1, \dots, m\}$:

$$Q^{\text{opt}} = q + \frac{1}{A_m}, \qquad w_i^{\text{opt}} = \frac{\tau_i}{A_m}. \tag{21}$$

6.4 Exact Infinite-Bagged Expression

For any fixed subset S, the embedded weight is $v_{S,i} = \mathbf{1}(i \in S)\,\tau_i/A_S$. Hence the infinite-bagged weight on forecaster i is

$$\bar{v}_{i,k} = E_S[v_{S,i}] = \tau_i\, E_R\left[\frac{k/m}{\tau_i + \sum_{j\in R}\tau_j}\right], \tag{22}$$

where R is drawn uniformly from all $(k-1)$-element subsets of $\{1, \dots, m\} \setminus \{i\}$. Since $\sum_i \bar{v}_{i,k} = 1$,

$$Q_k^{\text{bag},\infty} = q + \sum_{i=1}^{m} \frac{\bar{v}_{i,k}^2}{\tau_i}. \tag{23}$$

6.5 Excess Bagged Risk as a Weighted Squared Distance

Proposition 2 (Weight-deviation representation). *Let* $\Delta_{i,k} = \bar{v}_{i,k} - w_i^{\text{opt}}$. *Since both* $\bar{v}_k$ *and* w^{opt} *satisfy the adding-up constraint* $\sum_i \Delta_{i,k} = 0$, *the cross-term* $\sum_i \Delta_{i,k}/\tau_i = A_m^{-1} \sum_i \Delta_{i,k} = 0$. *Therefore*

$$R_k^{\text{bag},\infty} := Q_k^{\text{bag},\infty} - Q^{\text{opt}} = \sum_{i=1}^{m} \frac{(\bar{v}_{i,k} - w_i^{\text{opt}})^2}{\tau_i} \;\geq\; 0. \tag{24}$$

Proof. Expand $Q_k^{\text{bag},\infty} = q + \sum_i (w_i^{\text{opt}} + \Delta_{i,k})^2/\tau_i$. The linear term is $2\sum_i w_i^{\text{opt}} \Delta_{i,k}/\tau_i = (2/A_m)\sum_i \Delta_{i,k} = 0$. □

Remark 2. *The excess risk is a second-order (quadratic) quantity in the weight error. The vanishing of the linear term is not an approximation; it follows from* w^{opt} *satisfying the first-order condition of the constrained minimisation.*

Boundary conditions: $R_m^{\text{bag},\infty} = 0$ (at $k = m$, RSM recovers the full optimum) and $R_k^{\text{bag},\infty} \geq 0$ for all k, since $\tau_i > 0$.

7 Small-Heterogeneity Derivation

7.1 Assumption and Setup

Assumption 3 (Small heterogeneity). *Write* $\tau_i = \bar{\tau}(1 + \delta_i)$ *with* $m^{-1}\sum_i \delta_i = 0$ *and* $\max_i |\delta_i|$ *small. Define*

$$B = \frac{1}{\bar{\tau}}, \qquad V_\delta = \frac{1}{m}\sum_{i=1}^{m} \delta_i^2. \tag{25}$$

Under Assumption 2, $A_m = m\bar{\tau}$, $w_i^{\text{opt}} = (1+\delta_i)/m$, $Q^{\text{opt}} = q + B/m$.

7.2 Expansion of Average-Subset Risk

Write $A_S = \bar{\tau}(k + \Delta_S)$ where $\Delta_S = \sum_{i \in S} \delta_i$. Under simple random sampling without replacement,

$$E[\Delta_S] = 0, \qquad \operatorname{Var}(\Delta_S) = \frac{k(m-k)}{m-1} V_\delta. \tag{26}$$

A second-order Taylor expansion of $1/(k + \Delta_S)$ yields

$$Q_k^{\text{sub}} \approx q + \frac{B}{k} + B\frac{m-k}{k^2(m-1)} V_\delta. \tag{27}$$

The leading term B/k decays as $1/k$; the heterogeneity correction is smaller order. Defining $z_k = 1/k - 1/m$,

$$Q_k^{\text{sub}} - Q^{\text{opt}} \approx B\, z_k + B\frac{m-k}{k^2(m-1)} V_\delta. \tag{28}$$

7.3 Expansion of Bagged Weights

From (22), using $\tau_i = \bar{\tau}(1+\delta_i)$ and $\Delta_R = \sum_{j\in R}\delta_j$:

$$\bar{v}_{i,k} = \frac{1+\delta_i}{m} E_R\left[\frac{k}{k+\delta_i+\Delta_R}\right]. \tag{29}$$

Since $\sum_{j\neq i}\delta_j = -\delta_i$ (by $\sum_j \delta_j = 0$),

$$E_R[\Delta_R] = \frac{k-1}{m-1}(-\delta_i). \tag{30}$$

Expanding to first order in δ:

$$\bar{v}_{i,k} - w_i^{\text{opt}} = -\frac{\delta_i}{m}\cdot\frac{m-k}{k(m-1)} + O(\delta^2). \tag{31}$$

Since $(m-k)/(k(m-1)) = m/(m-1)\cdot z_k$, this can be written as

$$\Delta_{i,k} := \bar{v}_{i,k} - w_i^{\text{opt}} \approx -\frac{\delta_i}{m-1}\cdot z_k. \tag{32}$$

The weight deviation is *proportional to* $z_k = 1/k - 1/m$: it vanishes at $k = m$ (full pool) and grows as k decreases.

7.4 Endpoint-Corrected Approximation of Bagged Excess Risk

Inserting (32) into (24):

$$R_k^{\text{bag},\infty} = \sum_i \frac{\Delta_{i,k}^2}{\tau_i} \approx \frac{z_k^2}{(m-1)^2}\sum_i \frac{\delta_i^2}{\tau_i} = \frac{z_k^2}{(m-1)^2}\cdot B\sum_i \delta_i^2(1+O(\delta_i)) \approx \frac{BV_\delta m}{(m-1)^2}\, z_k^2. \tag{33}$$

Proposition 3 (Endpoint-corrected squared approximation)**.** *Under Assumptions 1–3,*

$$R_k^{\text{bag},\infty} \approx C_\delta\left(\frac{1}{k}-\frac{1}{m}\right)^2, \qquad C_\delta = \frac{BV_\delta}{m}\left(\frac{m}{m-1}\right)^2, \tag{34}$$

and the infinite-bagged risk is

$$Q_k^{\text{bag},\infty} \approx q + \frac{B}{m} + C_\delta\left(\frac{1}{k}-\frac{1}{m}\right)^2. \tag{35}$$

This approximation satisfies the exact endpoint $R_m^{\text{bag},\infty} = 0$*.*

Remark 3 (Comparison to C/k^2)**.** *For* $k \ll m$*,* $(1/k - 1/m)^2 \approx 1/k^2$*, so* (34) *reduces to the local form* C_δ/k^2*. But* (34) *is derived from the bagged weight expansion (valid for* *small k) rather than assumed, and it correctly sets $R_m = 0$. This makes it more reliable when the empirical optimal k is small relative to m.*

Remark 4 (Homoskedastic special case)**.** *If $\delta_i = 0$ for all i, then $V_\delta = C_\delta = 0$ and $Q_k^{\text{bag},\infty} = Q^{\text{opt}}$ for every k. Bagging fully removes the population dependence on k when all forecasters have equal precision.*

8 Finite Bagging under Small Heterogeneity

Combining Proposition 1 with the leading-order approximations $Q_k^{\text{sub}} - Q^{\text{opt}} \approx Bz_k$ and $R_k^{\text{bag},\infty} \approx C_\delta z_k^2$:

$$R_k^{\text{bag},G} = R_k^{\text{bag},\infty} + \frac{1}{G}(Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}) \approx \frac{B}{G}\, z_k + \left(1 - \frac{1}{G}\right) C_\delta\, z_k^2. \tag{36}$$

For compact notation set $a_G = B/G$ and $c_G = (1 - 1/G)C_\delta$. Then

$$Q_k^{\text{bag},G} \approx Q^{\text{opt}} + a_G\, z_k + c_G\, z_k^2. \tag{37}$$

Two limiting regimes:

- $G \to \infty$: the linear term vanishes ($a_G \to 0$), and only the squared term $C_\delta z_k^2$ remains. This is the infinite-bagged result.
- G small: the linear term $a_G z_k$ dominates for small k, pushing the optimal k toward a square-root rule.

The number of bags G therefore interpolates between the cube-root rule (large G) and the square-root rule (small G).

9 Optimal Subset Size

9.1 Average-Subset Optimum

Using the leading approximation $Q_k^{\text{sub}} \approx q + B/k$ and $A_0 = s + q$, the average-subset loss is $L_k^{\text{sub}} \approx (A_0 + B/k)(T-1)/(T-k)$. The first-order condition $\partial L_k^{\text{sub}}/\partial k = 0$ gives

$$A_0 k^2 + 2Bk - BT = 0, \tag{38}$$

with positive root

$$k^*_{\text{sub}} = \frac{-B + \sqrt{B^2 + A_0 BT}}{A_0} \sim \left(\frac{BT}{A_0}\right)^{1/2} \propto T^{1/2}. \tag{39}$$

The average-subset estimator has a *square-root optimal subset size.*

9.2 Infinite-Bagged Optimum

Using $L_k^{\text{bag},\infty} \approx (A + C_\delta z_k^2)(T-1)/(T-k)$ with $A = s + Q^{\text{opt}} = s + q + B/m$, the first-order condition is

$$Am^2k^3 + C_\delta k(m-k)^2 - 2C_\delta m(m-k)(T-k) = 0. \tag{40}$$

For $k \ll m$ this reduces to $Ak^3 + 3C_\delta k - 2C_\delta T \approx 0$, giving the *cube-root optimal subset size*:

$$k^*_{\text{bag},\infty} \approx \left(\frac{2C_\delta T}{A}\right)^{1/3} \propto \left(\frac{T}{m}\right)^{1/3}. \tag{41}$$

Substituting the full expression for C_δ:

$$k^*_{\text{bag},\infty} \sim \left[\frac{2BV_\delta T}{m(s+q+B/m)}\left(\frac{m}{m-1}\right)^2\right]^{1/3}. \tag{42}$$

Under stable heterogeneity and loss components, $k^* \propto (T/m)^{1/3}$.

9.3 Finite-Bagged Optimum

The finite-bagged loss $L_k^{\text{bag},G} \approx (A + a_G z_k + c_G z_k^2)(T-1)/(T-k)$ has first-order condition (for $k \ll m$):

$$Ak^3 + 2a_G k^2 + (3c_G - a_G T)k - 2c_G T = 0. \tag{43}$$

Two limiting regimes:

1. G large ($a_G \approx 0$): $k^*_{\text{bag},G} \approx (2c_G T/A)^{1/3}$, approaching the cube-root rule.
2. G small (first-order term dominates): $k^*_{\text{bag},G} \approx (a_G T/A)^{1/2} = (BT/(GA))^{1/2}$, approaching the square-root rule.

10 General Heterogeneity and the Power-Law Extension

10.1 Two Asymptotic Regimes

Proposition 4 (Small-heterogeneity regime). *Under Assumption 3 with* $\max_i |\delta_i|$ *small,* $\Delta_{i,k} = -(\delta_i/(m-1))z_k + O(\delta^2)$ *and* $R_k^{\text{bag},\infty} \approx C_\delta z_k^2$. *The OLS log-log slope is* $\hat{\alpha} = 2$ *exactly in the limit* $V_\delta \to 0$.

Proposition 5 (Large-heterogeneity regime). *Consider the dominant-forecaster model:* $\tau_1 = \tau_H$, $\tau_j = \tau_L$ *for* $j = 2, \dots, m$, *with* $\tau_H/\tau_L \to \infty$. *Then for each* $k \in \{1, \dots, m\}$*:*

$$\Delta_{1,k} \approx -\frac{m-k}{m}, \qquad \Delta_{j,k} \approx \frac{m-k}{m(m-1)}, \quad j \neq 1, \tag{44}$$

and

$$R_k^{\text{bag},\infty} \approx C'\left(1 - \frac{k}{m}\right)^2. \tag{45}$$

Proof. When $\tau_1 \to \infty$: forecaster 1 enters a random size-k subset with probability k/m; when included its weight $\to 1$. Hence $\bar{v}_{1,k} = k/m$ and $\Delta_{1,k} = k/m - 1 = -(m-k)/m$. For $j \neq 1$: forecaster j enters subsets not containing 1 with probability $k/m \cdot (m-k)/(m-1)$, receiving weight $\approx 1/k$ in those subsets. Thus $\bar{v}_{j,k} \approx (m-k)/(m(m-1))$ and $\Delta_{j,k} = (m-k)/(m(m-1))$ since $w_j^{\text{opt}} \to 0$. Inserting into (24) and letting $\tau_1 \to \infty$ gives (45). □

10.2 Log-Log Slope and Practical Interpolation

Writing $z_k = (m-k)/(km)$ and $u_k = \log(m-k)$, $v_k = -\log k$:

Proposition 6 (OLS slope in the two regimes)**.** *Let $\hat{u}, \hat{v}, V_u, V_v, C_{uv}$ denote sample mean, variance, and covariance of (u_k, v_k) over $k \in \{2, \ldots, m-1\}$.*

(i) Small het: $\log R_k = \text{const} + 2\log z_k$*, so* $\hat{\alpha} = 2$ *exactly.*

(ii) Large het: $\log R_k \approx \text{const} + 2u_k$*, and* $\hat{\alpha} = 2(V_u + C_{uv})/(V_u + 2C_{uv} + V_v)$*. When* $V_u = V_v$ *(symmetric range),* $\hat{\alpha} = 1$ *exactly.*

The key insight is that $V_u = V_v$ holds by symmetry of the range $k \in \{2, \ldots, m-1\}$ regardless of m, so $\hat{\alpha} \to 1$ as $\tau_H/\tau_L \to \infty$. For intermediate heterogeneity, $\hat{\alpha} \in (1, 2)$.

Remark 5 (Identification of $\hat{\alpha}$)**.** *The log-log slope $\hat{\alpha}$ is meaningfully identified only when C_δ is non-negligible, i.e. when the excess-risk curve $R_k^{\text{bag},\infty}$ has detectable curvature. When forecasters are near-homogeneous ($H \approx 1$), $C_\delta \approx 0$ and the curve is almost flat; in that regime $\hat{\alpha}$ is weakly identified and should not be interpreted as a structural heterogeneity parameter. Near-homogeneity is therefore characterised by $C_\delta \approx 0$, not by $\hat{\alpha} \approx 1$. Conditional on a non-negligible excess-risk curve, the small-heterogeneity expansion predicts $\hat{\alpha} = 2$ (Proposition 4), while the dominant-forecaster limit yields $\hat{\alpha} \approx 1$.*

10.3 Unified Power-Law and General k^*

Approximating $R_k^{\text{bag},\infty} \approx C z_k^{\alpha}$ for $\hat{\alpha} \in [1, 2]$, the total loss is

$$L_k^{\text{bag},\infty} \approx (A + C z_k^{\alpha}) \frac{T-1}{T-k}. \tag{46}$$

Using $dz/dk = -1/k^2$ and $k \ll m$, the interior first-order condition gives:

$$k^* \approx \left(\frac{C\alpha T}{A}\right)^{1/(\alpha+1)}. \tag{47}$$

The T-scaling exponent $1/(\alpha+1)$ ranges from $1/3$ (small het, $\alpha = 2$) to $1/2$ (dominant forecaster, $\alpha = 1$). The cube-root rule is the *lower bound* on the scaling exponent.

The three regimes are summarised in Table 1.

Table 1: Excess risk form and implied T-scaling under the fitted power law.

Regime	True form of R_k	C_δ	Effective $\hat{\alpha}$	k^* scaling
Near-homogeneous	≈ 0	≈ 0	weakly identified	any k near-optimal
Small het (identifiable)	$C_\delta z_k^2$	non-negligible	≈ 2	$T^{1/3}$
Intermediate	mixture	large	$\in (1, 2)$	$T^{1/(\alpha+1)}$
Dominant forecaster	$C'(1 - k/m)^2$	$\to \infty$	≈ 1	$T^{1/2}$

$\hat{\alpha}$ is interpretable only when C_δ is non-negligible (rows 2–4). Near-homogeneity means $C_\delta \approx 0$, not $\hat{\alpha} \approx 1$.

11 Dominance Regions

This section derives conditions under which RSM strictly beats the benchmarks of Section 3. Results are presented both exactly (using the common-loading identities) and under the small-heterogeneity approximation.

11.1 Setup

Define $A = s + Q^{\text{opt}}$ and $M_{\text{ave}} = s + Q^{\text{ave}}$, so $A \leq M_{\text{ave}}$ by optimality of w^{opt}. The comparisons require $T > m$ for the full optimal to be feasible, and $T > k$ for RSM.

11.2 Average-Subset RSM versus Full Estimated Optimal Weights

Proposition 7 (Exact average-subset dominance over L^{opt})**.** *Assume $T > m$. Average-subset RSM beats full estimated optimal weights at subset size k if and only if*

$$\big(Q_k^{\text{sub}} - Q^{\text{opt}}\big)(T - m) < A(m - k). \tag{48}$$

Proof. The average-subset loss is $L_k^{\text{sub}} = (s + Q_k^{\text{sub}})(T-1)/(T-k)$ and $L^{\text{opt}} = (s + Q^{\text{opt}})(T-1)/(T-m)$. Cancel $T - 1 > 0$, cross-multiply by $(T-k)(T-m) > 0$, write $s + Q_k^{\text{sub}} = A + (Q_k^{\text{sub}} - Q^{\text{opt}})$, and rearrange. □

Corollary 7.1 (Small-het approximation)**.** *Under Assumption 3 using $Q_k^{\text{sub}} - Q^{\text{opt}} \approx Bz_k = B/k - B/m$, condition* (48) *reduces to $k > B(T-m)/(mA)$.*

11.3 Average-Subset RSM versus Equal Weights

Proposition 8 (Exact average-subset dominance over L^{ave})**.** *Average-subset RSM beats equal weights at subset size k if and only if*

$$\left(s + Q_k^{\text{sub}}\right)(T-1) < M_{\text{ave}}(T-k). \tag{49}$$

Proof. From $L_k^{\text{sub}} < L^{\text{ave}}$ with $L^{\text{ave}} = M_{\text{ave}}$. □

Under the small-het approximation this gives a quadratic condition in k.

11.4 Infinite-Bagged RSM versus Full Estimated Optimal Weights

Proposition 9 (Exact infinite-bagged dominance over L^{opt})**.** *Infinite-bagged RSM beats full estimated optimal weights at subset size k iff*

$$R_k^{\text{bag},\infty}(T-m) < A(m-k), \tag{50}$$

where $R_k^{\text{bag},\infty}$ is the exact excess risk from (24)*.*

Corollary 9.1 (Small-het threshold)**.** *Under Assumption 3,* (50) *reduces to $Am^2k^2 > C_\delta(T-m)(m-k)$. The dominance threshold is*

$$\kappa^{\text{opt},\infty} = \frac{-C_\delta(T-m) + \sqrt{C_\delta^2(T-m)^2 + 4AC_\delta m^3(T-m)}}{2Am^2}. \tag{51}$$

Infinite-bagged RSM dominates full OLS for all $k \in (\kappa^{\text{opt},\infty}, m)$.

Remark 6 (Geometric interpretation)**.** *The right side $A(m-k)$ of* (50) *is the "estimation budget": it is large when $T \gg m$ (many data per parameter) or when $m-k$ is large (the subset uses far fewer forecasters than the full pool). As $k \to m$, both sides approach zero and the condition holds trivially (RSM equals full optimum in population at $k=m$).*

11.5 Finite-Bagged RSM versus Full Estimated Optimal Weights

Proposition 10 (Exact finite-bagged dominance over L^{opt})**.** *For G independent draws, finite-bagged RSM beats full estimated optimal weights at subset size k iff*

$$\left(R_k^{\text{bag},\infty} + \frac{Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}}{G}\right)(T-m) < A(m-k). \tag{52}$$

Under small heterogeneity, this reduces to $Am^2k^2 > (T-m)[a_G mk + c_G(m-k)^2/m^2 \cdot m^2]$, i.e. $Am^2k^2 > (T-m)[a_G mk + c_G(m-k)^2]$.

11.6 Infinite-Bagged RSM versus Equal Weights

Proposition 11 (Exact infinite-bagged dominance over L^{ave})**.** *Infinite-bagged RSM beats equal weights at subset size k iff*

$$(A + R_k^{\text{bag},\infty})(T-1) < M_{\text{ave}}(T-k). \tag{53}$$

Remark 7 (Sufficient conditions)**.** *Condition* (53) *is satisfied when: (i)* $R_k^{\text{bag},\infty}$ *is small (bagged weights close to oracle); and (ii)* $M_{\text{ave}} - A = Q^{\text{ave}} - Q^{\text{opt}}$ *is large (equal weights substantially worse than oracle). For large* T*, the leading constraint simplifies to* $Q_k^{\text{bag},\infty} < Q^{\text{ave}}$.

12 Exogenous Monte Carlo

12.1 Design

The exogenous simulation takes the true Σ_e as given and evaluates RSM under the common-loading model with known parameters.

Data-generating process. Forecast errors follow $u_{it} = B_{\text{scale}}F_t + H_{\text{scale}}\varepsilon_{it}$ where $F_t \sim N(0,1)$ (common factor) and $\varepsilon_{it} \sim N(0,1)$ (idiosyncratic). The parameter $B \in \{3, 8\}$ controls q (common loading) and $H \in \{1.05, 6\}$ controls the spread of idiosyncratic variances σ_i^2 (heterogeneity). Baseline: $m = 50$, $T = 240$.

Population objects. All theoretical quantities are computed analytically from Σ_e: Q^{opt}, C_δ, k^* from (41), $\hat{\alpha}$ from log-log OLS of $R_k^{\text{bag},\infty}$ on z_k, and exact dominance thresholds from (51). $Q_k^{\text{bag},\infty}$ is computed via (23) using $N_{\text{subsets}} = 2500$ Monte Carlo subset draws.

Simulation parameters. $N_{\text{rep}} = 500$, $G = 800$ bags, $N_{\text{subsets,pop}} = 2500$.

Objects reported. For each DGP we report: (i) the population k^* from grid search over $Q_k^{\text{bag},\infty}$; (ii) the empirical k^* minimising MSPE; (iii) the MSPE gain of RSM over EW; (iv) $\hat{\alpha}$ from log-log OLS on the population curve; (v) the mean relative error of the small-het approximation (34) versus the exact $R_k^{\text{bag},\infty}$.

12.2 Main Results ($m = 50$, $T = 240$)

Table 2 reports the four DGP combinations. The column "k^*(grid)" is the minimiser of $Q_k^{\text{bag},\infty}$ on the integer grid; "k^*(emp)" minimises the finite-sample MSPE over 500 replications.

Table 2: Exogenous MC: $m = 50$, $T = 240$, 500 replications, 800 bags.

B	H	k^*(grid)	k^*(emp)	Gain vs EW	$\hat{\alpha}$	R^2	Approx. err
3	1.05	1	3	0.1%	1.12	0.98	90%
8	1.05	1	3	0.1%	1.10	0.98	90%
3	6	3	6	12.7%	1.72	0.98	75%
8	3.5*	4	8	14.9%	1.70	0.99	77%

* Target $H = 6$; parameterisation yielded $H_{\text{actual}} = 3.51$. "Approx. err": mean $|R_k - C_\delta z_k^2|/R_k$ over all k. $\hat{\alpha}$: OLS slope from $\log R_k^{\text{bag},\infty} \sim \hat{\alpha} \log z_k$. At $H = 1.05$, $C_\delta \approx 0$ and the excess-risk curve is nearly flat; $\hat{\alpha}$ is weakly identified and should not be interpreted as a structural parameter in this row (see Remark on identification above).

Verification of theoretical predictions.

1. **Finite-bagging bridge**: Proposition 1 holds to machine precision ($< 9 \times 10^{-15}$) for all k and DGPs.

2. **Cube-root formula**: The formula (41) gives $k^*(\text{formula}) = k^*(\text{grid})$ exactly in all four DGPs. Despite approximation errors of 75–90%, the minimiser is correctly located because the error is approximately constant in k.

3. **Power-law shape**: $R^2 > 0.97$ for all DGP-power-law fits. At high heterogeneity ($H = 6$), $\hat{\alpha} \approx 1.7$, consistent with the intermediate-regime bounds in Proposition 6. At $H = 1.05$, $C_\delta \approx 0$ and the excess-risk curve is nearly flat; the fitted $\hat{\alpha} \approx 1.1$ is weakly identified and should not be read as a structural exponent (see identification remark in Section 10).

4. **$\hat{\alpha}$ driven by H, not B**: Both $B = 3$ and $B = 8$ give similar $\hat{\alpha}$ at the same H, confirming that $\hat{\alpha}$ reflects idiosyncratic precision heterogeneity, not the common-loading scale.

5. **Gains grow with H**: 0.1% at $H = 1.05$ vs 12–15% at $H = 3.5$–6. RSM provides most value when forecasters differ substantially.

6. **Dominance**: Applying (51), $\kappa^{\text{opt},\infty} < 1$ for all DGPs, meaning bagged RSM dominates full estimated OLS at $k = 1$. Empirically this is confirmed: $\min \text{MSPE(RSM)} < \text{MSPE(full OLS)}$.

12.3 T-Scaling: Cube-Root Law

Table 3 varies $T \in \{80, 240, 640, 1000\}$ at $B = 3$, $H \approx 1.05$. The continuous cube-root formula value (41), the integer grid minimiser of $Q_k^{\text{bag},\infty}$, and the empirical optimum are reported.

At $H = 1.05$, the excess risk is essentially zero at every k, so the loss surface is flat and the population optimum is $k^* = 1$ for all T studied. The cube-root formula correctly

Table 3: Exogenous T-scaling: $B = 3$, $H \approx 1.05$, $m = 50$.

T	k^*(formula, cont.)	k^*(grid, bag$^\infty$)	k^*(emp)	Gain
80	0.57	1	2	0.02%
240	0.82	1	2	0.02%
640	1.13	1	4	0.06%
1000	1.31	1	3	0.03%

predicts this: the continuous value grows from 0.57 to 1.31 (confirming $k^* = 1$ as the rounded optimum). Gains are $< 0.1\%$ regardless of T, making the scaling law empirically invisible.

Interpretation. The cube-root law is detectable only when heterogeneity is large enough to create meaningful excess risk. The $H = 6$ baseline confirms this: the formula gives $k^* = 3$ and $k^* = 4$ for $B = 3$ and $B = 8$ respectively, matching the grid search exactly. The T-scaling experiment at $H = 1.05$ documents the flat regime, which is relevant to the SPF application (Section 15).

13 Motivation for the Endogenous Case

The exogenous theory treats Σ_e as known. In practice, forecast-error covariances must be estimated from data. This introduces an additional source of noise that the exogenous analysis does not capture.

Two additional challenges arise:

1. **Covariance estimation error**: estimating Σ_e from m series of length T requires $m(m+1)/2$ parameters. When m/T is not negligible, the sample covariance is a poor estimate of Σ_e, and combination weights computed from it can be far from the true optimal.

2. **Non-common-loading structure**: real forecast-error covariances are not exactly common-loading. We study two alternatives—Toeplitz (exponential decay) and factor-structured Σ_e—to assess robustness.

The endogenous experiment asks whether the qualitative predictions of the exogenous theory (interior optimal k, RSM dominance over EW and full OLS, $T^{1/3}$ scaling) survive in a realistic setting where Σ_e is estimated and may not follow the common-loading structure.

14 Endogenous Monte Carlo

14.1 Design

Covariance structures. Two Σ_e designs are used:

- **Toeplitz**: $(\Sigma_e)_{ij} = 0.8^{|i-j|}$ (exponential decay with $\rho = 0.8$).
- **Factor**: $\Sigma_e = \Lambda\Lambda' + \Psi$ with $r = 2$ random factors, factor strength 0.5, idiosyncratic variance 0.25.

Coefficient designs.

- **B2** (dense): $\beta_j = 1/m$ for all j (equal signal per forecaster).
- **B4** (sparse): non-zero on a small subset, larger values; concentrates predictive power.

Simulation parameters. Baseline: $m = 50$, $T = 200$, $N_{\text{rep}} = 200$, $K_{\text{RS}} = 200$ bags, OOS length 30. T-scaling: $T \in \{100, 150, 200, 300, 500, 800\}$, $N_{\text{rep}} = 120$, $K_{\text{RS}} = 150$.

RSM implementation. At each OOS period, the subset minimum-variance weights are computed as

$$\hat{w}_{i,S} = \frac{(\hat{\Sigma}_{e,S}^{-1}\iota_S)_i}{\iota_S'\hat{\Sigma}_{e,S}^{-1}\iota_S}, \tag{54}$$

where $\hat{\Sigma}_{e,S}$ is the $k \times k$ principal submatrix of the rolling training-window sample covariance matrix $\hat{\Sigma}_e$. This matches Definition 1 with Σ_e replaced by its estimate; under the common-loading structure these weights reduce to $\hat{\tau}_i/\hat{A}_S$, but for general Toeplitz or factor Σ_e the full inverse-covariance formula is used. Population objects $Q_k^{\text{bag},\infty}$ and $R_k^{\text{bag},\infty}$ are estimated by Monte Carlo using the true Σ_e (for population benchmarking) and the full empirical procedure (for the general $\hat{\Sigma}_e$).

14.2 Main Results ($m = 50$, $T = 200$)

Key findings.

1. **Interior optimum in all scenarios.** The RMSFE-optimal k^* is 11–14, confirming a genuine interior trade-off between approximation bias and estimation noise.
2. **Dominance.** $\kappa_{\text{exact}} = 2$ in all four scenarios: bagged RSM beats full estimated OLS at every $k \geq 2$.

Table 4: Endogenous MC: $m = 50$, $T = 200$, 200 replications, 200 bags.

Σ_e	β	k^*(RMSFE)	$k^*(L_{\text{bag}})$	RMSFE gain	κ_{exact}	Q^{opt}
Toeplitz	B2	11	32	3.5%	2	0.155
Toeplitz	B4	11	26	9.8%	2	0.155
Factor	B2	14	29	5.0%	2	0.005
Factor	B4	14	29	6.1%	2	0.004

"RMSFE gain": $(L^{\text{EW}} - L^{\text{RSM},k^*})/L^{\text{EW}}$. κ_{exact}: smallest k at which RSM beats full estimated OLS (evaluated on the empirical RMSFE curve). $k^*(L_{\text{bag}})$: minimiser of the estimated bagged loss including inflation.

3. **Sparsity amplifies gains.** B4 (sparse) yields larger RMSFE gains than B2 (dense) for both Toeplitz and Factor designs (9.8% vs 3.5% for Toeplitz; 6.1% vs 5.0% for Factor). Sparse signal concentrates predictive power, increasing effective heterogeneity.

4. **Factor structure shifts k^* rightward.** $k^* = 14$ for Factor vs $k^* = 11$ for Toeplitz. The Factor DGP has lower Q^{opt} (0.004–0.005 vs 0.155), meaning higher signal-to-noise ratio and a larger optimal subset to average out idiosyncratic noise.

5. **Gap between $k^*(L_{\text{bag}})$ and k^*(RMSFE).** The inflation-based criterion $k^*(L_{\text{bag}}) =$ 26–32 substantially exceeds the OOS RMSFE optimum $k^* =$ 11–14. This gap illustrates the limits of the maintained inflation approximation $I(T,k) = (T-1)/(T-k)$: it was derived for a single constrained OLS estimator with $k-1$ free parameters, whereas bagging over many overlapping subsets reduces the effective degrees of freedom below $k-1$. Additionally, the formula does not account for the estimation of Σ_e from $m = 50$ series. The cube-root rule should therefore be read as a theoretical guide under the maintained approximation, not as a precise predictor of the RMSFE-optimal k.

14.3 T-Scaling in the Endogenous Case

Table 5: Endogenous T-scaling: $m = 50$, B4, 120 replications.

Σ_e	$T = 100$	$T = 150$	$T = 200$	$T = 300$	$T = 500$	$T = 800$
Toeplitz	10	10	14	14	18	22
Factor	14	14	14	18	26	26

Log-log OLS regressions of k^* on T give slopes of 0.394 (Toeplitz, $p = 0.001$, $R^2 = 0.94$) and 0.367 (Factor, $p = 0.006$, $R^2 = 0.88$). Both are consistent with the theoretical $T^{1/3}$ prediction and reject $T^{1/2}$ at the 10% level.

14.4 Why the Endogenous k^* Differs from the Exogenous Prediction

Table 6: Comparison of k^*(RMSFE) across settings.

Setting	T	k^*
Exog. MC, $H = 1.05$ (population, any B)	240	1
Exog. MC, $H = 6$, $B = 3$	240	3–6
Endo. MC, Toeplitz, B4	200	11
Endo. MC, Factor, B4	200	14
GDP bivariate (full sample)	≈ 100	19
GDP bivariate (no-COVID)	≈ 100	21

The endogenous k^* is substantially higher than the exogenous population prediction for two structural reasons:

1. **Covariance estimation noise**: in the endogenous case, $\hat{\Sigma}_e$ is estimated from $m = 50$ series of length $T = 200$. The Kan–Zhou (2007) penalty for covariance estimation is $(T - 2)/(T - m - 2) = 198/148 \approx 1.34$, implying a 34% upward shift in the inflation-adjusted optimal k. This explains much of the gap between the exogenous population optimum ($k^* = 2$) and the endogenous RMSFE optimum ($k^* = 11$–14).

2. **Non-common-loading Σ_e**: neither Toeplitz nor Factor satisfies the common-loading assumption. The bagged weight formula $w_{i,S} = \hat{\tau}_i/\hat{A}_S$ is suboptimal for these structures, introducing additional approximation error not captured by the theory.

15 Empirical Application: SPF (Near-Optimal Case)

15.1 Motivation

The exogenous theory predicts that gains are negligible when forecasters are near-homogeneous ($V_\delta \approx 0$, $H \approx 1$): the excess risk curve $R_k^{\text{bag},\infty} \approx C_\delta z_k^2$ is flat, and the equal-weighted mean is near-optimal. The Survey of Professional Forecasters (SPF) provides a natural test: professional economists with similar information sets and training constitute a near-homogeneous panel.

15.2 Data and Setup

- **Source**: Philadelphia Fed SPF, Q1 1985–Q4 2024.
- **Panel**: $N_t \approx 30$–53 respondents per round (median 36 for real GDP, 35 for CPI).

- **Individual forecasts**: each respondent's own submission (no additional regression estimation step).
- **RSM**: at each round, draw k forecasters without replacement and take the simple average of their forecasts. $G = 1000$ draws. Sweep $k \in \{1, \ldots, k_{\max}\}$ where $k_{\max}$ is the largest k at which $\geq 80\%$ of rounds have $N_t \geq k$. Since no estimated precision weights are used, infinite bagging mechanically recovers the full equal-weighted mean: $E_S\left[\frac{1}{k}\sum_{i\in S} f_i\right] = \frac{1}{N}\sum_{i=1}^{N} f_i$. The SPF section therefore functions as a sanity check—confirming that when the method collapses toward equal averaging and forecaster heterogeneity is limited, it is hard to beat the mean—rather than as evidence for the weighted-RSM theory.
- **Target variables**: RGDP growth ($h = 1$ and $h = 4$); CPI inflation ($h = 1$ and $h = 4$). Actual values from FRED.

15.3 Results: Hard to Beat the Mean

Table 7: SPF RSM vs equal-weighted mean; OOS Q1 1985–Q4 2024.

Series	h	Med. N	k^*	$k = \sqrt{N}$	RMSFE (Mean)	RMSFE (RSM)	Gain	
RGDP	1	36	3	6	6.405	6.392	+0.2%	
RGDP	4	36	13	6	2.208	2.205	+0.1%	No
CPI	1	35	10	6	0.974	0.969	+0.5%	
CPI	4	35	3	6	1.471	1.470	+0.1%	

DM test significant at 10% level. Gain = $(L^{\mathrm{EW}} - L^{\mathrm{RSM}})/L^{\mathrm{EW}}$; positive means RSM is better. Gains are economically negligible, consistent with the sanity-check interpretation (see text).

Gains are uniformly below 0.5% and statistically insignificant. Three compounding mechanisms explain why:

1. **Near-homogeneous precision.** Professional economists use similar models and information sets, placing this application in the $H \approx 1.05$ regime. The MC confirms that gains are $< 0.1\%$ there.

2. **No additional estimation noise.** SPF submissions carry no OLS estimation error, removing the channel through which RSM gains arise in the GDP application.

3. **Small panel.** At $N \approx 36$, the combinatorial diversity of RSM is limited. The gains from heterogeneous precision grow with m (Proposition 3).

Interpretation. The SPF results confirm the sanity check: when RSM uses equal within-subset averaging and the infinite-bagged limit mechanically recovers the equal-weighted mean, a finite-G approximation offers no reliable advantage. Gains are below 0.5% and statistically indistinguishable from zero in all cases. This is consistent with the theory's prediction: near-homogeneous forecasters ($C_\delta \approx 0$) imply a flat excess-risk curve, and RSM reduces to equal weighting. The SPF does not constitute evidence for or against the weighted-RSM theory; it confirms the expected limiting behaviour.

16 Empirical Application: FRED-MD GDP Nowcasting

16.1 Data and Setup

- **Source**: 102 real-time quarterly vintages of FRED-MD (September 1999 – March 2025; Q4 2003 excluded for missing data).
- **Sample**: estimation from approximately 1970Q1 to the forecast origin, giving $T_{\text{eff}} \approx$ 79–183 quarters (median $\approx$ 100). Predictor pool: $m \approx 126$ monthly FRED-MD series aggregated to quarterly frequency.
- **Target**: log-differenced real GDP growth at one quarter ahead.

Individual forecasts (bivariate). For predictor j and time t:

$$y_t = \alpha_j + \beta_j x_{j,t} + \varepsilon_t. \tag{55}$$

Each of the $m \approx 126$ predictors yields one individual forecast per period.

Individual forecasts (AR+X). Following Elliott et al. (2013):

$$y_t = \alpha_j + \sum_{p=1}^{P} \varphi_p y_{t-p} + \beta_j x_{j,t} + \varepsilon_t, \tag{56}$$

where $P \in \{0, 1, 2, 3, 4\}$ is chosen by AIC at each vintage.

RSM combination. At each period, RSM samples k individual forecasts and computes sum-to-one constrained OLS combination weights. $G = 1000$ draws for the main estimate. Robustness: sweep $k \in \{2, \ldots, 55\}$ with $G = 200$. Default $k = \lfloor\sqrt{T}\rfloor \approx$ 9–14.

Benchmarks. Equal-weighted mean, Lasso, Ridge, Random Forest.

16.2 Bivariate Results

RMSFE by k: fixed-rule performance and oracle diagnostic. The primary out-of-sample evidence uses the fixed rule $k = \lfloor\sqrt{T}\rfloor \approx 9$, chosen before observing the evaluation-sample outcomes. Tables 8 and 9 report this fixed-k performance.

Table 8: FRED-MD GDP bivariate RSM: full sample (102 quarters).

Model	MAFE	Sig.	RMSFE	Sig.
Mean	0.54		1.10	
RSM	0.45	**	0.76	*
Lasso	0.51		0.73	*
Ridge	0.49	*	0.98	
Random Forest	0.55		1.24	

$k = \lfloor\sqrt{T}\rfloor \approx 9$. $^*p < 0.10$; $^{**}p < 0.05$ (one-sided DM test vs Mean).

Table 9: FRED-MD GDP bivariate RSM: excluding COVID-19 (Q2–Q3 2020).

Model	MAFE	Sig.	RMSFE	Sig.
Mean	0.42		0.60	
RSM	0.38	**	0.53	**
Lasso	0.45		0.58	
Ridge	0.39	**	0.54	**
Random Forest	0.41		0.57	*

As a descriptive diagnostic, we also sweep $k \in \{2, \ldots, 55\}$ and examine the shape of the RMSFE(k) curve *ex post*. The curve is U-shaped in both samples, confirming the interior optimum predicted by the theory. The ex-post minimisers $k^* = 19$ (full) and $k^* = 21$ (no-COVID) are *oracle values* selected on the evaluation sample and should not be interpreted as genuine out-of-sample evidence:

- Full sample: oracle $k^* = 19$, $\min \text{RMSFE} = 0.662$; fixed rule $k = 9$: RMSFE $= 0.762$.
- No-COVID: oracle $k^* = 21$, $\min \text{RMSFE} = 0.499$; fixed rule $k = 9$: RMSFE $= 0.529$.

The gap between the fixed-rule and oracle RMSFE indicates that a data-driven k selection rule (e.g. rolling validation) could deliver further gains, but establishing this is left for future work.

16.3 AR+X Results

Table 10: FRED-MD GDP AR+X RSM: full sample.

Model	MAFE	Sig.	RMSFE	Sig.
Mean	0.56		1.27	
RSM (AR+X)	0.45	**	0.76	*
Lasso	0.52		0.77	
Ridge	0.50	**	1.02	*
Random Forest	0.57		1.40	

Table 11: FRED-MD GDP AR+X RSM: excluding COVID-19.

Model	MAFE	Sig.	RMSFE	Sig.
Mean	0.41		0.57	
RSM (AR+X)	0.39	*	0.52	*
Lasso	0.47		0.63	
Ridge	0.39	**	0.54	**
Random Forest	0.40		0.57	

Bivariate vs AR+X.

	Bivariate		AR+X	
Sample	k^*	minRMSFE	k^*	minRMSFE
Full	19	0.662	19	**0.617**
No-COVID	21	**0.499**	16	0.506

Adding AR lags reduces the full-sample RMSFE by 6.8%. The no-COVID AR+X $k^* = 16$ is closer to the endogenous MC range 11–14 than the bivariate $k^* = 21$.

16.4 Connection to Theory

The empirical $k^* = 16$–21 at $T \approx 100$ substantially exceeds the exogenous population prediction ($k^* \approx 1$–5) for two reasons:

1. **Underpowered covariance estimation**: $m \approx 126 > T \approx 100$ places the application in the $m/T > 1$ regime where the Kan–Zhou correction diverges. The standard inflation formula (13) accounts for estimating k combination weights but not for estimating Σ_e from $m > T$ series. This creates substantially more noise at small k, pushing k^* rightward.

2. **Individual forecast quality**: each predictor's individual forecast is an estimated OLS regression with its own estimation error. This adds a layer of noise absent in the endogenous MC (where individual signals are drawn from a clean DGP), further pushing k^* upward.

The AR+X specification partially resolves the second issue: it improves individual forecast quality, bringing the no-COVID k^* from 21 to 16 (closer to the endogenous MC range). The gap that remains is attributable to the $m/T > 1$ regime.

17 Summary and Conclusion

17.1 Summary of Theoretical Results

Under the common-loading model (Assumption 2):

1. **Exact weight-deviation formula** (Proposition 2): $R_k^{\text{bag},\infty} = \sum_i (\bar{v}_{i,k} - w_i^{\text{opt}})^2/\tau_i$ is the fundamental object governing the trade-off in k.

2. **Endpoint-corrected approximation** (Proposition 3): under small heterogeneity, $R_k^{\text{bag},\infty} \approx C_\delta(1/k - 1/m)^2$, derived from the bagged weight expansion, not assumed.

3. **Cube-root law**: under the maintained inflation approximation, minimising the inflation-adjusted loss (14) yields $k^* \propto T^{1/3}$, a lower scaling exponent than the square-root rule of the average-subset estimator.

4. **Finite bagging** (Proposition 1): with G draws, the excess risk is $R_k^{\text{bag},G} \approx (B/G)z_k + (1 - 1/G)C_\delta z_k^2$. A large number of bags is needed to recover the cube-root law; for small G the rule transitions toward the square-root.

5. **General heterogeneity** (Propositions 4–6): the excess risk follows a power law $R_k \approx C z_k^\alpha$ with $\alpha \in [1, 2]$, implying $k^* \propto T^{1/(\alpha+1)} \in [T^{1/3}, T^{1/2}]$.

6. **Exact dominance** (Propositions 9–**??**): closed-form conditions determine when RSM beats full estimated OLS and equal weights. Under small heterogeneity these yield explicit lower-bound thresholds on k.

17.2 Summary of Monte Carlo Results

Exogenous MC. All six theoretical predictions are verified. The finite-bagging bridge holds to machine precision. The cube-root formula locates k^* exactly in all DGPs despite 75–90% approximation error in R_k levels. The power-law slope $\hat{\alpha} \in [1.1, 1.7]$ increases with heterogeneity. Dominance over full estimated OLS holds at $k = 1$ for all DGPs.

Endogenous MC. The qualitative structure survives under estimated covariance and non-common-loading structures: an interior optimum exists, RSM dominates full estimated OLS at $k = 2$, sparsity amplifies gains (up to 9.8%), and T-scaling exponents of 0.39 (Toeplitz) and 0.37 (Factor) are consistent with $T^{1/3}$ under the extended estimation-noise interpretation. The maintained inflation approximation overpredicts the RMSFE-optimal k (26–32 vs 11–14), reflecting that bagged RSM has lower effective degrees of freedom than a single size-k constrained OLS.

17.3 Summary of Empirical Results

SPF. Gains are $< 0.5\%$ and statistically insignificant across all four variable/horizon combinations. This confirms the theory's prediction for the near-homogeneous regime: when forecasters have similar precision ($H \approx 1$), the equal-weighted mean is nearly optimal and hard to beat.

FRED-MD GDP. RSM yields statistically significant RMSFE reductions of 18–30% over the equal-weighted mean (full sample), confirming that large heterogeneous predictor panels ($m \approx 126$, $\alpha \approx 1.7$) are precisely the environment where RSM is most valuable. The U-shaped RMSFE(k) curve and interior optimum $k^* = 16$–21 match the endogenous MC qualitatively. The AR+X specification improves individual forecast quality and brings k^* closer to the MC benchmark.

17.4 Conclusion

This paper establishes RSM as a theoretically grounded and empirically effective method for forecast combination. Its main advantage over equal weighting is largest when forecasters are heterogeneous in precision. Near-homogeneity is characterised by $C_\delta \approx 0$ (a flat excess-risk curve), not by a specific value of the power-law slope $\hat{\alpha}$, which is weakly identified when the curve is flat. When C_δ is non-negligible and heterogeneity is high (as in FRED-MD GDP), RSM delivers 10–30% RMSFE reductions; when $C_\delta \approx 0$ (as in the SPF), RSM collapses toward equal weighting and gains are negligible. A practitioner can estimate $\hat{\alpha}$ by log-log regression of $R_k^{\text{bag},\infty}$ on z_k over a sparse grid (Remark 4 of the addendum), then set $k^* = \lfloor (\hat{C}\hat{\alpha}T/A)^{1/(\hat{\alpha}+1)} \rfloor$ and verify by evaluating L_k at $k^* \pm 2$.

A Exact Dominance Table

Table 12 collects exact and small-het dominance conditions. Let $\mathcal{E}_k = Q_k^{\text{sub}} - Q^{\text{opt}}$, $z_k = 1/k - 1/m$, $\Delta_k = Q_k^{\text{sub}} - Q_k^{\text{bag},\infty} \geq 0$.

Table 12: Exact dominance conditions and small-het approximations.

Comparison	Exact condition	Small-het approximation
Avg-sub vs L^{opt}	$\mathcal{E}_k(T-m) < A(m-k)$	$k > B(T-m)/(mA)$
Avg-sub vs L^{ave}	$(s+Q_k^{\text{sub}})(T-1) < M_{\text{ave}}(T-k)$	quadratic in k
∞-bag vs L^{opt}	$R_k^{\text{bag},\infty}(T-m) < A(m-k)$	$Am^2k^2 > C_\delta(T-m)(m-k)$
Finite-bag vs L^{opt}	$(R_k^{\text{bag},\infty} + \Delta_k/G)(T-m) < A(m-k)$	$Am^2k^2 > (T-m)[a_G mk + c_G(m-k)^2]$
∞-bag vs L^{ave}	$(A + R_k^{\text{bag},\infty})(T-1) < M_{\text{ave}}(T-k)$	cubic in k

$A = s + Q^{\text{opt}}$; $M_{\text{ave}} = s + Q^{\text{ave}}$; $a_G = B/G$; $c_G = (1-1/G)C_\delta$.

B Data Sources

- **FRED-MD**: Real-time monthly vintages, September 1999–March 2025. 103 vintages; Q4 2003 excluded. McCracken and Ng (2016).
- **FRED-QD**: Quarterly vintages matched to monthly by vintage date.
- **SPF**: Philadelphia Fed Survey of Professional Forecasters. RGDP (columns RGDP1–RGDP5) and CPI (columns CPI1–CPI5). Actuals: GDPC1 and CPIAUCSL from FRED (downloaded June 2026).

C Software

All simulations in R ($\geq$ 4.2). Key packages: `dplyr`, `glmnet`, `ranger`, `forecast`, `readxl`. All random seeds fixed. Scripts:

- Exogenous MC: `rsm_common_loading_grid_weights..._final.R`
- Endogenous MC: `endogenous_rsm_toeplitz_factor.R`
- GDP: `GDP_constrained/GDP_all_k_results.R` and `GDP_ar_bivariate_allk.R`
- SPF: `GDP_constrained/SPF_RSM_analysis.R`

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